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A Defensive Framework Against Adversarial Attacks on Machine Learning-Based Network Intrusion Detection Systems

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***Abstract*—As cyberattacks become increasingly sophisticated, advanced Network Intrusion Detection Systems (NIDS) are critical for modern network security. Traditional signature-based NIDS are inadequate against zero-day and evolving attacks. In response, machine learning (ML)-based NIDS have emerged as promising solutions; however, they are vulnerable to adversarial evasion attacks that subtly manipulate network traffic to bypass detection. To address this vulnerability, we propose a novel defensive framework that enhances the robustness of ML-based NIDS by simultaneously integrating adversarial training, dataset balancing techniques, advanced feature engineering, ensemble learning, and extensive model fine-tuning. We validate our framework using the NSL-KDD and UNSW-NB15 datasets. Experimental results show, on average, a 35% increase in detection accuracy and a 12.5% reduction in false positives compared to baseline models, particularly under adversarial conditions. The proposed defense against adversarial attacks significantly advances the practical deployment of robust ML-based NIDS in real-world networks.**

***Index Terms*—Adversarial Attacks, Network Intrusion Detection Systems (NIDS), Machine Learning, Adversarial Training, Ensemble Learning, Feature Engineering, Network Protocols, Cybersecurity**

I. Introduction

The increasing volume and sophistication of cyberattacks pose a significant challenge to the modern network security. As organizations increasingly rely on interconnected systems, including IoT technologies, the need for effective cybersecurity measures has become more urgent. Among the various tools used to safeguard these networks, NIDS play a critical role in identifying and mitigating potential malicious activities. Traditionally, these systems have been based on signature detection methods, where predefined rules and patterns are used to identify known attacks. While these approaches are effective for detecting well-documented threats, they struggle with identifying novel or rapidly evolving zero-day attacks—exploits that leverage previously unknown vulnerabilities.

To overcome the limitations of signature-based NIDS, the focus has shifted more towards ML-based anomaly detection. ML-based NIDS analyze network traffic behavior and use statistical models to detect deviations from established norms, which allows them to flag suspicious activity. These systems hold the potential to detect previously unseen attacks by learning the underlying behavior of benign network traffic. However, despite their advantages, ML-based NIDS are extremely vulnerable to adversarial attacks. In such attacks, adversaries manipulate network traffic in subtle ways to deceive the ML models, causing them to misclassify malicious activity as benign. These adversarial attacks have become a growing concern because they exploit weaknesses in the model's decision boundaries, allowing attackers to bypass detection.

A key aspect of adversarial attacks on NIDS involves manipulating mutable network features—such as packet sizes, network timings, and flow patterns. These features can often be altered by attackers in an attempt to evade detection, as small changes can make malicious traffic appear normal to a ML model. However, in real-world networks, attackers typically do not have full control over certain features like timing or routing behaviors due to network infrastructure constraints and the nature of shared resources. Additionally, mutable features are often interdependent, meaning that changing one feature, such as packet size, may indirectly affect others, like timing or flow patterns, because of the underlying structure of network protocols. These constraints are frequently overlooked in current research, which tends to assume that attackers can manipulate features freely [1], [2].

Another significant gap in the existing research on ML-based NIDS is the lack of robust defense mechanisms to protect against adversarial attacks. Most studies have either focused on improving detection accuracy under non-adversarial conditions [3]–[6] or on generating adversarial attacks to evaluate vulnerabilities in these systems [7]–[9]. This gap underscores the need for stronger, more resilient defenses that can effectively mitigate the risks posed by adversarial threats in real-world network environments.

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針對基於機器學習的網路入侵偵測系統的對抗性攻擊的防禦框架

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摘要—隨著網路攻擊日益複雜，先進的網路入侵偵測系統（NIDS）對現代網路安全至關重要。傳統基於簽名的NIDS對零日攻擊和演變中的攻擊無法有效應對。為此，基於機器學習（ML）的NIDS作為有前景的解決方案應運而生；然而，它們容易受到對抗性逃逸攻擊的威脅，這些攻擊會巧妙地操縱網路流量以繞過偵測。為了解決這一脆弱性，我們提出了一個新的防禦框架，通過同時整合對抗性訓練、數據集平衡技術、先進的特徵工程、集成學習和廣泛的模型微調來增強基於ML的NIDS的穩健性。我們使用NSL-KDD和UNSW-NB15數據集驗證了我們的框架。實驗結果顯示，與基線模型相比，平均偵測準確率提高了35%，偽陽性率降低了12.5%，尤其是在對抗性條件下。我們提出的對抗性攻擊的防禦顯著推進了穩健的基於ML的NIDS在實際網路中的應用部署。索引詞—對抗性攻擊，網路入侵偵測系統（NIDS），機器學習，對抗性訓練，集成學習，特徵工程，網路協議，網路安全

I. 前言

網路攻擊的數量與複雜度日益增加，對現代網路安全構成嚴峻挑戰。隨著組織越來越依賴互連系統，包括IoT 技術，有效網路安全的必要性變得更加緊迫。在保護這些網路的各種工具中，NIDS 在識別和緩解潛在惡意活動方面扮演關鍵角色。傳統上，這些系統基於簽名偵測方法，使用預先定義的規則和模式來識別已知攻擊。雖然這些方法對偵測有文獻記錄的威脅有效，但它們難以識別新出現或快速演變的零日攻擊——利用先前未知漏洞的漏洞利用。

為克服基於簽名的 NIDS 的限制，重點已轉向基於機器學習的異常偵測。基於機器學習的 NIDS 分析網路流量行為，並使用統計模型偵測與既定常規的偏差，這使它們能夠標記可疑活動。這些系統有潛力通過學習良性網路流量的基本行為來偵測先前未見過的攻擊。然而，儘管它們有優點，基於機器學習的 NIDS 對於敵對攻擊極其脆弱。在這種攻擊中，攻擊者以微妙的方式操縱網路流量來欺騙機器學習模型，導致它們將惡意活動錯誤分類為良性。這些敵對攻擊已成為一個日益擔憂的問題，因為它們利用模型決策邊界的弱點，使攻擊者能夠繞過偵測。

對於網路入侵偵測系統（NIDS）的對抗性攻擊而言，操控可變的網路特徵是一個關鍵面向—例如封包大小、網路時間和流量模式。這些特徵通常可被攻擊者修改，試圖逃避偵測，因為微小的變更能使惡意流量對機器學習模型看起來正常。然而，在實際網路環境中，攻擊者通常無法完全控制某些特徵，如時間或路由行為，這是因為網路基礎設施的限制以及共享資源的性質。此外，可變特徵之間常互相依存，意味著改變其中一個特徵，例如封包大小，可能因網路協議的底層結構，間接影響其他特徵，如時間或流量模式。這些限制在當前研究中常被忽略，研究往往假設攻擊者可以自由操控特徵 [1], [2]。

目前針對基於機器學習的網路入侵偵測系統（ML-NIDS）的研究中，另一個顯著的差距在於缺乏能夠有效防禦對抗性攻擊的堅固防禦機制。大多數研究或許專注於在非對抗性條件下提升偵測準確度 [3]–[6] ，或許專注於產生對抗性攻擊以評估這些系統的弱點 [7]–[9]。這個差距突顯了需要更強、更具韌性的防禦機制，以有效降低真實網路環境中對抗性威脅所帶來的風險。

In this paper, we propose a novel defensive framework to enhance the security and robustness of ML-based NIDS against adversarial attacks by integrating key strategies: adversarial training, dataset balancing, advanced feature engineering, ensemble learning, and extensive model fine-tuning. Adversarial training, which exposes ML models to both normal and adversarial samples, improves their ability to detect manipulated inputs. To address dataset imbalance, we apply tailored oversampling techniques that mitigate bias toward benign traffic. Advanced feature engineering focuses on key traffic features and their interactions, making it harder for attackers to manipulate them without detection. Ensemble learning, combining multiple ML models, strengthens robustness by reducing reliance on any single model. Finally, extensive model fine-tuning optimizes detection accuracy while minimizing false positives and ensuring low-latency performance with GPU-based parallel computation.

To evaluate the effectiveness of our framework, we conducted experiments using two well-established datasets in the field of NIDS: NSL-KDD [10] and UNSW-NB15 [11]. Our experimental results demonstrate that the proposed framework significantly improves detection accuracy while also reducing the rates of false positives and false negatives, especially in the presence of adversarial attacks. This substantial performance gain illustrates the importance of integrating adversarial training, feature engineering, ensemble learning, and model fine-tuning into ML-based NIDS.

Contributions. The contributions are summarized as follows:

- We propose a novel adversarial attack generation methodology that ensures attacks are both realistic and practical. This is achieved by introducing perturbations that comply with network protocols, preserve feature interdependencies, and maintain attack feasibility within real-world network environments. The use of a genetic algorithm ensures that the generated adversarial examples remain both stealthy and effective, making them challenging for ML-based NIDS to detect.
- We introduce a novel defensive framework to improve the resilience of ML-based NIDS. The framework integrates multiple strategies, including adversarial training, protocol-aware feature engineering, balancing techniques, and extensive hyperparameter tuning. By incorporating domain-specific knowledge of network traffic and protocol standards, the framework strengthens the model's ability to detect adversarial manipulations that exploit weaknesses in ML models.
- We test our framework using two widely recognized benchmark datasets, NSL-KDD and UNSW-NB15, to evaluate performance under both regular and adversarial scenarios. Specifically, our framework achieves a 35% improvement in detection accuracy and reduces the false positive rate by 12.5% on average compared to baseline models, particularly when adversarial attacks are present.

The rest of this paper is organized as follows: Section II explores ML-based NIDS and adversarial attacks. Section III reviews related works in the field. Section IV provides a detailed explanation of how we mount our adversarial attack on ML models, focusing specifically on its impact on ML-based NIDS. Section V details the core defensive strategies used in this work, including adversarial training, dataset balancing, advanced feature engineering, ensemble modeling, and hyperparameter tuning. Section VI presents the experimental results, highlighting performance metrics and insights into model robustness. Finally, Section VII concludes the paper.

II. Background Information

A. ML-based NIDS

ML-based NIDS detect malicious activities by identifying anomalies in network traffic patterns. Unlike traditional signature-based systems, ML-based NIDS use historical data to learn and detect both known and unknown threats, such as zero-day attacks. Key ML models include supervised techniques like Decision Trees, Random Forests, SVMs, and Logistic Regression, as well as unsupervised approaches like k-Means Clustering and Autoencoders. Deep learning models, including CNNs, RNNs, and LSTMs, excel in processing high-dimensional and sequential data, offering improved detection for complex attack patterns and anomalies [12], [13], [16]–[18].

B. Overview of Adversarial Attacks on ML-based NIDS

Adversarial attacks on ML-based NIDS involve subtle modifications to input data that mislead the ML model into misclassifying malicious traffic as benign. Formally, adversaries introduce a perturbation $\boldsymbol{\delta}$ to the input feature vector $\mathbf{x}$ to produce $\hat{\mathbf{x}}$, where:

$$\hat{\mathbf{x}} = \mathbf{x} + \boldsymbol{\delta}, \quad \text{and} \quad f(\mathbf{x}) \neq f(\hat{\mathbf{x}})$$

The perturbations must meet constraints such as minimal magnitude, feature specificity, and protocol compliance to remain undetected:

- **Minimal Perturbation:** Ensure $\|\boldsymbol{\delta}\|_p \leq \epsilon$ to avoid detection.
- **Feature-Specific Constraints:** Continuous features are perturbed, while categorical features remain immutable.
- **Protocol Compliance:** Modified traffic adheres to standards like TCP, UDP, and ICMP.

Adversaries solve an optimization problem to minimize $\|\boldsymbol{\delta}\|_p$ while ensuring $\hat{\mathbf{x}} \in C$, the set of valid inputs adhering to protocol standards.

C. Flows, Packets, and ML-Based Pattern Detection

Network traffic analysis involves packets aggregated into flows. Packets, the basic transmission units, are analyzed for header anomalies or unauthorized protocol usage. However, flows offer a richer context by tracking sequences of packets with shared attributes, such as IP addresses and transport protocols. This allows detection of more sophisticated attacks like SYN flooding, incomplete handshakes, or session hijacking.

ML models analyze flows to identify deviations in patterns, such as abnormal flow frequencies (indicating DDoS attacks) or flow size variations (suggesting data exfiltration). Models

在本文中，我們提出一個新的防禦框架，透過整合關鍵策略來提升基於機器學習的NIDS對抗性攻擊的安全性與韌性：對抗性訓練、數據集平衡、先進的特徵工程、集成學習以及廣泛的模型微調。對抗性訓練透過讓機器學習模型接觸正常與對抗性樣本，提升其偵測被操弄輸入的能力。為了處理數據集不平衡問題，我們應用定制的過度採樣技術，減少對良性流量偏見。先進的特徵工程專注於關鍵流量特徵及其交互作用，讓攻擊者更難在不被偵測的情況下操弄它們。集成學習透過結合多個機器學習模型，減少對單一模型的依賴，從而增強韌性。最後，廣泛的模型微調優化偵測準確度，同時最小化偽陽性，並透過基於GPU的平行計算確保低延遲性能。

為了評估我們架構的有效性，我們使用 NIDS 領域中兩個已建立的數據集（NSL-KDD [10] 和 UNSW-NB 15 [11]）進行了實驗。實驗結果顯示，所提出的架構顯著提高了偵測準確率，同時也降低了假陽性和假陰性的率，尤其是在存在對抗性攻擊的情況下。這顯著的性能提升說明了整合對抗性訓練、特徵工程、集成學習和模型微調至基於 ML 的 NIDS 的重要性。

貢獻。貢獻總結如下：

- 我們提出了一種新的對抗性攻擊生成方法論，確保攻擊既真實又實用。
這是通過引入符合網絡協議、保留特徵相互依存關係並在真實網絡環境中維持攻擊可行性來實現的。使用遺傳算法確保生成的對抗性範例既隱蔽又有效，使其對基於 ML 的 NIDS 來說難以偵測。
- 我們引入了一個新的防禦架構來提高基於 ML 的 NIDS 的韌性。該架構整合了多種策略，包括對抗性訓練、協議感知特徵工程、平衡技術和廣泛的超參數調整。通過結合網絡流量和協議標準的特定領域知識，該架構增強了模型偵測利用 ML 模型弱點進行對抗性操縱的能力。
- 我們使用兩個廣泛認可的基準數據集，NSL-KDD 和 UNSW-NB15，來測試我們的框架，以評估在常規和對抗性場景下的性能。具體來說，我們的框架在基線模型上平均實現了 35% 的檢測精度提升，並將假陽性率降低了 12.5%，尤其是在存在對抗性攻擊時。

本文餘下的內容分為以下幾部分：第二節探討基於機器學習的網路入侵偵測系統（NIDS）與對抗性攻擊。第三節回顧該領域的相關研究。第四節詳細說明我們如何針對機器學習模型進行對抗性攻擊。

特別強調其對基於機器學習的NIDS的影響。第五節詳細介紹本研究使用的核心防禦策略，包括對抗性訓練、數據集平衡、先進的特徵工程、集成建模以及超參數調優。第六節呈現實驗結果，突出性能指標以及模型穩健性的洞見。最後，第七節總結全文。

II. 背景資訊

A. 基於機器學習的網路入侵偵測系統

基於機器的學習網路入侵檢測系統 (ML-based NIDS) 通過識別網絡流量模式中的異常來檢測惡意活動。與傳統的基於簽名的系統不同，基於機器的學習網路入侵檢測系統使用歷史數據來學習和檢測已知和未知威脅，例如零日攻擊。關鍵的機器學習模型包括監督技術，如決策樹、隨機森林、SVM 和邏輯回歸，以及非監督方法，如 k-均值聚類和自編碼器。深度學習模型，包括 CNN、RNN 和 LSTM，在處理高維度和序列數據方面表現出色，為複雜的攻擊模式和異常提供更優的檢測效果 [12], [13], [16]–[18]。

B. 基於機器學習的網路入侵偵測系統（NIDS）的對抗性攻擊概述

針對基於機器學習的網路入侵偵測系統（NIDS）的對抗性攻擊，涉及輕微修改輸入數據，使機器學習模型誤將惡意流量分類為良性。正式來說，攻擊者向輸入特徵向量 x 引入一個擾動 $\boldsymbol{\delta}$ ，以產生 ˆx，其中：

$$\hat{\mathbf{x}} = \mathbf{x} + \boldsymbol{\delta}, \quad \text{and} \quad f(\mathbf{x}) \neq f(\hat{\mathbf{x}})$$

這些擾動必須滿足最小幅度、特徵特定性以及協議合規性等約束，以保持不被偵測：

- 最小擾動：確保 $\|\boldsymbol{\delta}\|_p \leq \epsilon$ 以避免偵測。
- 功能特定限制：連續特徵會被干擾，而類別特徵則保持不變。
- 協議符合性：修改後的流量遵循 TCP、UDP 和 ICMP 等標準。

威脅者解決一個優化問題，以最小化 $\|\boldsymbol{\delta}\|_p$ ，同時確保 ˆx $\in C$，即符合協議標準的有效輸入集合。

C. 流量、封包和基於機器學習的模式偵測

網路流量分析涉及將封包聚合為流量。封包是基本傳輸單位，會被分析標頭異常或未授權協議使用。然而，流量能提供更豐富的上下文，透過追蹤具有共同屬性（如 IP 位址和傳輸協議）的封包序列，這允許偵測更複雜的攻擊，例如 SYN 洪水、不完整的握手或會話劫持。

機器學習模型分析流量以識別模式中的偏差，例如異常的流量頻率（表示DDoS攻擊）或流量大小變化（暗示數據洩露）。模型

like RNNs and LSTMs are particularly effective for sequential flow analysis, enabling the detection of coordinated or multi-stage attacks. By grouping flows, NIDS can uncover patterns spanning multiple connections, such as botnet activity or port scanning. Behavioral profiling further enhances detection by learning normal communication patterns and flagging deviations.

TABLE I
COMPARISON OF ANALYSIS LEVELS

Level of Analysis	Description	Useful For Detecting
Single Packets	Analysis of individual packets, focusing on headers and payload.	Header anomalies, unauthorized protocol use, malformed packets.
Flows	Sequence of packets with shared attributes (e.g., source/destination IP, protocol).	SYN flooding, incomplete handshakes, slow-rate DDoS.
Groups of Flows	Multiple flows between the same source and destination.	Coordinated attacks, multi-stage intrusions, port scanning.

III. Related Works

Adversarial attacks on ML-based NIDS have garnered significant attention due to the increasing reliance on ML to detect sophisticated cyber threats. Recent studies have focused on system vulnerabilities and strategies to mitigate these attacks.

A. Proposed Defensive Mechanisms for ML-based NIDS

Wang et al. [19] and Reza et al. [20] explore robust defenses for ML-based NIDS, such as data augmentation, robustness certification, and protocol-aware feature manipulation, to address adversarial attacks and distribution shifts. These works emphasize the importance of feature engineering and real-world constraints in improving NIDS robustness. Building on these approaches, our framework incorporates adversarial training, balancing techniques, and advanced feature engineering to address distribution shifts and enhance resilience against a broader range of attacks.

B. Adversarial Attack Generation and Evasion Techniques

Peng et al. [21] highlight how small perturbations can undermine DoS detection and propose enhancements to boundary-based methods to improve resilience against adversarial inputs. Zhang et al. [24] investigate Universal Adversarial Perturbations (UAP) on DNN-based NIDS, demonstrating significant attack success rates and underscoring the need for robust defensive architectures. While these studies focus on adversarial example generation and exposing vulnerabilities, our work emphasizes realistic attack generation under real-world constraints, ensuring protocol compliance and feature interdependencies. Additionally, we propose robust defenses to counter these attacks effectively.

C. Attack Transferability Across Machine Learning Models

Piplai et al. [23] examine attack transferability using GAN-generated adversarial examples that bypass standard NIDS and suggest ensemble defenses to combat such threats. Expanding on this concept, our framework incorporates ensemble models in the fitness function for attack generation, demonstrating how transferable attacks can be mitigated through integrated defensive strategies. This approach strengthens system robustness and reduces evasion risks across classifiers.

IV. Adversarial Attack Generation Methodology

Existing methods for generating adversarial examples for ML-based NIDS often overlook real-world constraints, such as protocol compliance, feature mutability, and interdependencies. Many assume attackers can freely manipulate features or test attacks under unrealistic conditions, such as post-feature engineering perturbations or testing on the same classifiers used for training. These limitations often lead to biased results, reducing their practical applicability.

To address these issues, we propose a realistic attack generation framework using a genetic algorithm optimized for feature mutability, interdependence, and network constraints. This approach generates adversarial traffic that evades ML-based NIDS in operational environments. The algorithm employs an ensemble classifier as a black-box baseline, simulating real-world conditions and exploring attack portability across unseen models. The ensemble includes Logistic Regression, SVM, Decision Trees, and Random Forest to ensure robust evaluations.

Initialization. The algorithm starts with a population of adversarial instances, each representing perturbed attack examples targeting mutable features (e.g., `duration`, `src_bytes`, `dst_bytes`). Perturbations dynamically adapt to feature correlations, ensuring plausible network traffic patterns. For example, increasing `src_bytes` proportionally adjusts the permissible range for `dst_bytes` to maintain consistency.

Fitness Function. Adversarial instances are evaluated using a multi-objective fitness function balancing evasion effectiveness and perturbation magnitude:

$$f(\hat{x}) = P(\text{benign} \mid \hat{x}) - \lambda \cdot \|x - \hat{x}\|$$

Here, $P(\text{benign} \mid \hat{x})$ measures the likelihood of the instance being classified as benign, while λ penalizes large deviations from the original input. This approach ensures minimal yet effective perturbations while maintaining realistic traffic patterns.

Selection and Reproduction. Top-performing individuals are selected via tournament selection and recombined through crossover (80% probability) to inherit both feature values and correlation-aware perturbation ranges. This process enhances the realism and stealth of adversarial instances by respecting feature interdependencies.

Mutation. A 1% mutation rate introduces random changes to features while dynamically adjusting correlated feature ranges. For example, increasing `src_bytes` during mutation

如RNN和LSTM對於序列流量分析特別有效，能夠偵測協調性或多階段的攻擊。透過分組流量，NIDS可以發現跨越多個連接的模式，例如僵尸網活動或端口掃描。行為分析進一步增強偵測能力，透過學習正常通信模式並標記偏差。

表 I
分析層級比較

分析層級	描述	適用於偵測
單個封包	個別封包分析，重點在於標頭與資料負載。	標頭異常，未授權協議使用，錯誤格式封包。
流量	具有共同屬性的封包序列（例如，來源/目標 IP、協議）。	SYN 洪水攻擊，不完整的握手，低速率 DDoS。
流量群組	多個流量來自同一來源與目標地。	協調式攻擊、多階段入侵、端口掃描。

III. 相關研究

針對基於機器學習的網路入侵偵測系統的對抗性攻擊，由於機器學習在偵測複雜網路威脅上的依賴度日益增加，已引起廣泛關注。近年來的研究重點集中在系統漏洞以及減緩這些攻擊的策略上。

A. 基於機器學習的網路入侵偵測系統的防禦機制建議

王等人 [19] 與瑞薩等人 [20] 探索基於機器學習的網路入侵偵測系統 (NIDS) 的堅韌防禦措施，例如數據增強、堅韌性認證以及協議感知的特徵操作，以應對對抗性攻擊和分佈變動。這些研究強調特徵工程和實際世界限制在提高 NIDS 堅韌性方面的重要性。基於這些方法，我們的框架整合了對抗性訓練、平衡技術和先進的特徵工程，以應對分佈變動並增強對更廣泛攻擊範圍的韌性。

B. 對抗性攻擊生成與迴避技術

Peng 等人 [21] 強調微小干擾如何削弱 DoS 檢測，並提出針對基於邊界的方法的改進，以增強對抗性輸入的韌性。Zhang 等人 [24] 調查基於 DNN 的 NIDS 上的通用對抗性干擾 (UAP)，證明顯著的攻擊成功率和強調了對堅固防禦架構的需求。雖然這些研究專注於對抗性範例生成和暴露漏洞，我們的工作強調在真實世界約束下生成真實的攻擊，確保協議符合性和功能相互依存性。此外，我們提出堅固的防禦措施來有效對抗這些攻擊。

C. 攻擊在機器學習模型間的傳遞性

Piplai 等人 [23] 檢討攻擊可遷移性，使用 GAN 生成的對抗性範例來繞過標準 NIDS，並建議集成防禦來對抗這種威脅。擴展這個概念，我們的框架在攻擊生成中將集成模型納入適應函數，說明了可遷移攻擊如何能通過集成防禦策略來減輕。這種方法增強了系統韌性並降低了跨分類器的迴避風險。

IV. 對抗性攻擊生成方法學

現有生成基於機器學習的 NIDS 對抗性範例的方法常忽略實際世界限制，例如協議合規性、特徵可變性和相互依存性。許多假設攻擊者可以自由操縱特徵或在非現實的條件下測試攻擊，例如特徵工程後的擾動或測試用於訓練的相同分類器。這些限制常導致偏差結果，降低其實際應用性。

為了解決這些問題，我們提出了一個使用優化後的基因演算法的實際攻擊生成框架，該演算法專注於特徵可變性、相互依存性和網絡限制。這種方法生成能夠在實際操作環境中繞過基於機器學習的網絡入侵檢測系統（NIDS）的對抗流量。該演算法採用集成分類器作為黑盒基準，模擬真實世界情況並探索在未見過的模型之間的攻擊可移植性。集成包括邏輯回歸、支持向量機（SVM）、決策樹和隨機森林，以確保穩健的評估。

初始化。該演算法從一組對抗實例開始，每個實例代表針對可變特徵（例如，持續時間、來源_字節、目的_字節）的變動攻擊範例。變動動態適應特徵相關性，確保合理的網絡流量模式。例如，增加來源_字節會按比例調整目的_字節的可接受範圍以保持一致性。適應度函數。對抗實例使用平衡逃逸有效性和變動幅度多目標適應度函數進行評估：

$$f(\hat{x}) = P(\text{benign} \mid \hat{x}) - \lambda \cdot \|x - \hat{x}\|$$

在這裡， P（良性 | $\hat{}x$）衡量實例被分類為良性的可能性，而 λ 會懲罰與原始輸入的較大偏差。這種方法確保了最小但有效的變動，同時保持了真實的流量模式。

選擇與繁衍。表現優異的個體透過競賽選擇被選出，並透過交配（80%機率）重新組合，以繼承特徵值與相關感知的干擾範圍。這個過程透過尊重特徵間的相互依存關係，提升了對抗實例的真實性與隱蔽性。

基因突變。1%的突變率會隨機改變特徵，同時動態調整相關特徵的範圍。例如，在突變期間增加 src_bytes

recalibrates the permissible range for `dst_bytes`, ensuring changes remain plausible and undetectable.

Iteration and Termination. The algorithm iterates through selection, crossover, and mutation for up to 1000 generations or until the fitness function converges. Elitism ensures the top 5% of adversarial instances are retained across generations. This dynamic perturbation system continuously adapts to feature interdependencies, enabling the generation of adversarial examples that effectively evade detection while mimicking legitimate network traffic.

By integrating correlation-aware perturbations and real-world constraints, our framework produces stealthy and effective adversarial instances. These examples align with network traffic patterns, enhancing their ability to bypass NIDS in operational settings.

V. Building Robust NIDS against Adversarial Attacks

Building a resilient NIDS against adversarial attacks requires a multi-layered defense strategy that integrates adversarial training, robust feature engineering, and careful model fine-tuning. Adversarial training strengthens models by exposing them to adversarial examples, improving their ability to detect manipulations. Feature engineering enhances resilience by focusing on high-quality, attack-resistant features, while model fine-tuning ensures classifiers are adaptable and optimized for real-world environments. These combined defenses fortify the system against diverse attack vectors and enhance overall performance.

A. Adversarial Training

Adversarial training is a widely recognized defense mechanism for mitigating adversarial attacks. This method trains the model on both normal network traffic and adversarial examples designed to deceive the system, enabling it to better identify and classify maliciously altered inputs. By incorporating adversarial samples, the NIDS becomes more resilient to subtle perturbations and manipulation.

However, generating realistic adversarial examples that simulate real-world attacks remains a key challenge. If adversarial samples do not reflect actual attack patterns, the model risks overfitting to synthetic data, impairing its ability to generalize to unseen scenarios. This can reduce effectiveness in detecting both normal and slightly perturbed malicious traffic, thereby compromising robustness.

To overcome these challenges, we incorporated Monte Carlo perturbations during adversarial training, targeting mutable features while maintaining protocol compliance. This approach generated realistic adversarial examples, exposing the model to diverse, real-world scenarios and improving its resilience to evasion attacks. Additionally, we applied the Synthetic Minority Over-sampling Technique (SMOTE) to balance the dataset by oversampling minority classes, ensuring adequate representation of both adversarial and benign traffic. Combining SMOTE with adversarial training enhanced the model's performance and generalizability, enabling it to detect underrepresented attack types without being biased by imbalanced data.

B. Feature Extraction, Selection, and Engineering

Feature extraction, selection, and engineering are vital for constructing robust ML-based NIDS that can withstand adversarial attacks. These processes enable the system to focus on the most relevant features, reducing the attack surface and improving interpretability. Carefully engineered features are also more resistant to adversarial manipulation, making it harder for attackers to exploit vulnerabilities.

1) Feature Extraction and Selection: Feature extraction transforms raw network traffic data into structured information usable by ML models. Key data points include packet sizes, timing, protocol types, and header fields, with protocol-specific features being particularly critical due to unique vulnerabilities in protocols like TCP, UDP, and ICMP. For example, features like SYN-ACK sequences or retransmission rates in TCP can reveal manipulation attempts.

Feature selection identifies the most informative attributes for the model, such as traffic flow statistics, protocol usage metrics, and anomaly indicators. This step enhances detection efficiency, reduces susceptibility to irrelevant data, and improves generalization across attack types and network environments.

2) Feature Engineering for Security: Feature engineering complements extraction and selection by creating domain-specific features that enhance the model's ability to detect adversarial attacks. A key consideration is distinguishing between mutable features, like packet size and timing, which attackers can alter without breaking protocols, and immutable features, such as certain protocol flags, which are harder to manipulate without triggering alarms or disruptions.

Adversarial perturbations of mutable features are constrained by protocol limitations. Excessive changes to packet sizes or flow rates can disrupt connections, making attacks detectable. In bidirectional protocols like TCP, attackers must manipulate both communication sides without breaking the connection, increasing detection likelihood.

Our approach emphasizes protocol-aware feature engineering, ensuring feature dependencies are respected. For example, perturbing TCP flags must adhere to protocol rules; violations would result in unrealistic traffic patterns. Leveraging protocol-specific knowledge improves detection accuracy and minimizes false positives.

Additionally, feature engineering aggregates traffic data over time windows or multiple segments, helping the model detect subtle anomalies undetectable in single-packet analyses. For instance, tracking statistical properties (e.g., mean, variance) of network flows over time reveals long-term attack patterns, slow-moving anomalies, or multi-stage intrusions. By continuously refining protocol-aware features, the NIDS remains effective across a broad spectrum of adversarial techniques.

會重新校準 dst_bytes 的可接受範圍，確保變化保持合理且無法偵測。
迭代與終止。該演算法透過選擇、交配和變異進行迭代，最多1000代或直到適應度函數收斂。精英主義確保頂尖5%的對抗實例能在各世代中得以保留。這套動態干擾系統持續適應特徵間的相依關係，能產生能有效避開偵測的同時模擬合法網路流量對抗實例。

透過整合相關感知干擾和真實世界限制，我們的框架能產生隱蔽且有效的對抗實例。這些實例符合網路流量模式，增強其能在實際作業環境中繞過NIDS的能力。

V. 建立抗對抗攻擊的堅固NIDS

建立能抗對抗攻擊的堅固NIDS需要整合多層次防禦策略，包括對抗訓練、堅固的特徵工程和謹慎的模型微調。對抗訓練透過讓模型接觸對抗實例來強化其能力，提升偵測操弄的能力。特徵工程透過專注於高品質、抗攻擊的特徵來增強韌性，而模型微調則確保分類器能適應並優化於真實世界環境。這些整合的防禦措施加強了系統對各種攻擊向量 的防禦，並提升整體性能。

A. 對抗訓練

對抗訓練是一種廣泛認可的防禦機制，用於緩解對抗攻擊。這種方法在正常網絡流量和設計用於欺騙系統的對抗示例上訓練模型，使其能夠更好地識別和分類惡意修改的輸入。通過加入對抗樣本，NIDS 對於微小的干擾和操縱變得更加強韌。

然而，生成模擬真實世界攻擊的現實對抗示例仍然是一個關鍵挑戰。如果對抗樣本沒有反映實際的攻擊模式，模型風險過度適應合成數據，損害其泛化到未見情況的能力。這可能降低檢測正常和略微干擾的惡意流量的有效性，從而犧牲其穩健性。

為了克服這些挑戰，我們在對抗訓練期間加入了蒙地卡羅干擾，針對可變特徵進行處理，同時保持協議的兼容性。這種方法生成了真實的對抗樣本，讓模型接觸到多樣化的真實世界情境，並提高其對迴避攻擊的韌性。此外，我們應用了合成少數過度採樣技術（SMOTE）來平衡數據集，透過過度採樣少數類別，確保對抗流量和良性流量的代表性。結合SMOTE與對抗訓練增強了模型的效能與泛化能力，使其能夠偵測到未充分表達的攻擊類型，而不受不平衡數據的影響。

B. 特徵提取、選擇和工程

特徵提取、選擇和工程對於構建能夠承受對抗攻擊的穩健基於 ML 的 NIDS 至關重要。這些過程使系統能夠專注於最相關的特徵，減少攻擊面並提高可解釋性。經過精心設計的特徵也對於對抗操縱更加抵禦，使攻擊者更難利用漏洞。

1) 特徵萃取與選擇：特徵萃取將原始網路流量數據轉換為可供機器學習模型使用的結構化資訊。關鍵數據點包括封包大小、時間、協議類型及標頭欄位，其中特定協議的特徵尤為關鍵，因為像 TCP、UDP 和 ICMP 等協議存在獨特的弱點。例如，TCP 中的 SYN-ACK 序列或重傳率等特徵可以揭露操弄嘗試。

特徵選擇識別對模型最有資訊的屬性，例如流量統計資料、協議使用指標及異常指標。這個步驟提升偵測效率，降低對不相關數據的敏感性，並改善跨不同攻擊類型及網路環境的泛化能力。

2) 基於安全的特徵工程：特徵工程
complements 抽取與選擇，透過建立專域
提升模型偵測對抗性攻擊能力的具體特徵。一個重要的考量是區分可變特徵（例如封包大小和時間），攻擊者可以在不破壞協議的情況下修改這些特徵，以及不可變特徵（例如某些協議旗標），這些特徵更難以操縱，且無法觸發警報或中斷。

可變特徵的對抗干擾受到協議限制。封包大小或流量速率的過度變化可能中斷連接，使攻擊可被偵測。在像 TCP 這樣的雙向協議中，攻擊者必須在不中斷連接的情況下操控雙方通信，增加偵測的可能性。

我們的方法強調協議感知的特徵工程，確保特徵相依性得到尊重。例如，干擾 TCP 標誌必須遵守協議規則；違反規則會導致不切實際的流量模式。利用協議特定知識可提高偵測準確性並最小化偽陽性。

此外，特徵工程會匯總流量資料於時間窗格或多個區段，協助模型偵測單一封包分析中無法偵測的微妙異常。例如，追蹤網路流量在時間上的統計特性（例如平均值、變異數），可揭示長期攻擊模式、緩慢發生的異常或分階段入侵。透過持續精煉協議感知特徵，NIDS 能在廣泛的敵對技術範圍內保持有效。

C. Model Construction and Ensemble Learning

For model construction, we employ an ensemble of classifiers, including traditional models such as Logistic Regression and Random Forest, as well as deep learning models like LSTM and MLP. The ensemble approach improves robustness by combining the strengths of multiple models, ensuring that the system is not reliant on any single model's weaknesses. The fitness of each model is evaluated using both normal and adversarial traffic, ensuring that the NIDS remains effective in real-world scenarios.

VI. Evaluation

We evaluated our proposed defensive framework using two widely recognized NIDS benchmark datasets: NSL-KDD [10] and UNSW-NB15 [11]. The NSL-KDD dataset, an improved version of KDD Cup 1999, addresses issues like redundant records and data imbalance and includes four attack types: Denial of Service (DoS), Probe, User to Root (U2R), and Remote to Local (R2L). UNSW-NB15 represents modern network traffic with diverse attack types such as Fuzzers and Backdoors. Created using the IXIA PerfectStorm tool, it includes 49 features, providing a comprehensive evaluation benchmark.

Our evaluation assessed the defensive framework across configurations combining strategies such as *Adversarial Training*, *Balancing*, *Feature Preprocessing*, *Feature Engineering*, and *Model Fine-Tuning*. Models were tested on two datasets: *regular testing data* (unaltered traffic) and *adversarial testing data* (perturbed traffic designed to evade detection).

A. Evaluation Metrics

We evaluated each model using the following metrics:

- **Accuracy**: The proportion of correctly classified instances.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

- **Precision**: The proportion of true positives out of the predicted positives.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

- **Recall**: The proportion of true positives out of the actual positives.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

B. Classifier Ensembles and Improvement Process

We evaluate two classifier ensembles: the *Traditional Classifier Ensemble* and the *Deep Learning Ensemble*. The evaluation process is cumulative, beginning with a baseline, followed by the inclusion of additional defensive strategies. The improvement process proceeds as follows:

- **Baseline**: Initial model evaluation with no defensive strategies.
- **Adversarial Training & Balancing**: Applying adversarial training and balancing (e.g., SMOTE).
- **Feature Engineering & Preprocessing**: Adding feature engineering and preprocessing techniques.
- **Model Fine-Tuning**: Applying final model fine-tuning for optimal performance.

Each classifier ensemble is evaluated on both the *NSL-KDD* and *UNSW-NB15* datasets, with results for both *normal testing datasets* (unaltered traffic) and *adversarial testing datasets* (perturbed traffic).

C. Performance on NSL-KDD and UNSW-NB15 Datasets

Tables II and III summarize the performance of the Traditional Classifier Ensemble (TC) and Deep Learning Ensemble (DC) on the *NSL-KDD* and *UNSW-NB15* datasets. Results are reported for both normal and adversarial testing datasets using metrics such as *Accuracy*, *Precision*, and *Recall* at each stage of the cumulative improvement process.

Configurations with adversarial training consistently improved robustness against adversarial inputs by reducing false negatives, though they introduced a slight increase in false positives. Balancing techniques like SMOTE mitigated this trade-off, particularly in high-dimensional datasets such as *UNSW-NB15*, where false positive rates would have been higher without balancing.

Adversarial training was most effective on adversarial datasets but had limited impact on unaltered datasets. In these cases, SMOTE directly improved recall and reduced missed detections. Preprocessing steps such as scaling and feature interaction further enhanced model performance, while advanced feature engineering improved generalization, especially for complex classifiers.

Fine-tuning provided incremental performance gains, particularly in tree-based and deep learning models, by optimizing performance for nuanced scenarios. Together, these strategies significantly improved accuracy and recall across all evaluated configurations.

VII. Conclusion

In this paper, we proposed a comprehensive defensive framework to enhance the resilience of machine learning-based NIDS against adversarial attacks by integrating five key defensive strategies: *Adversarial Training*, *Balancing*, *Feature Preprocessing*, *Feature Aggregation/Engineering*, and *Model Fine-Tuning*. A critical aspect of our approach involved generating realistic adversarial samples that adhered to network protocol constraints and feature interdependencies. These samples were not only used to evaluate the framework's robustness but also to refine it through adversarial training, enabling the models to adapt to challenging attack scenarios. This interplay between adversarial sample generation and defense strategies ensured that the framework effectively mitigates evolving threats, making it practical for real-world deployment.

References

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C. 模型建構與集成學習

在模型建構上，我們採用分類器集成，包括傳統模型如邏輯回歸和隨機森林，以及深度學習模型如 LSTM 和 MLP。集成方法透過結合多個模型的優勢來提升穩健性，確保系統不依賴任何單一模型的弱點。每個模型的適合度是透過正常和敵對流量來評估，確保 NIDS 在實際場景中保持有效。

VI. 評估

我們使用兩個廣泛認可的NIDS基準資料集評估我們所提出的防禦框架：NSL-KDD [10]與UNSW-NB15 [11]。NSL-KDD資料集是KDD Cup 1999的改進版本，解決了重複記錄和資料不平衡等問題，包含四種攻擊類型：服務拒絕（DoS）、探查、使用者到根（U2R）和遠端到本地（R2L）。UNSW-NB15代表現代網路流量，包含多樣的攻擊類型，如模糊器（Fuzzers）和後門（Backdoors）。使用IXIA PerfectStorm工具創建，包含49個特徵，提供全面的評估基準。

我們的評估在結合對抗訓練、平衡、特徵預處理、特徵工程和模型微調等策略的配置下進行，評估防禦框架。模型在兩個資料集上進行測試：常規測試資料（未修改的流量）和對抗測試資料（設計用於規避偵測的干擾流量）。

A. 評估指標

我們使用以下指標評估每個模型：

- 準確率：正確分類的比例。
  立場。

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

- 精確率：在預測的正例中，真實正例的比例。
  正例。

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

- 請記得：實際正確的陽性數比例
  正面。

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

B. 分類器集成與改進流程

我們評估兩個分類器集成：傳統分類器集成和深度學習集成。評估過程是累積性的，從基準線開始，然後加入額外的防禦策略。改進過程如下：

- 基準值：無防禦策略的初始模型評估。
- 對抗訓練與平衡：應用對抗訓練與平衡（例如，SMOTE）。
- 特徵工程與預處理：增加特徵
  工程與預處理技術。
- 模型微調：應用最終模型微調以達到最佳性能。

每個分類器集成都在 NSL-KDD 上進行評估，以及UNSW-NB15數據集，結果包含正常測試數據集（未修改流量）和對抗測試數據集（干擾流量）。

C. 在 NSL-KDD 和 UNSW-NB15 資料集上的性能

表 II 和表 III 概括了傳統分類器集成 (TC) 和深度學習集成 (DC) 在 NSL-KDD 和 UNSW-NB15 資料集上的性能。結果報告了在累積改進過程的每個階段，使用準確率、精確率和召回率等指標對正常和對抗測試資料集進行評估。

具有對抗訓練的配置通過減少偽陰性結果，持續提高對抗輸入的穩定性，但引入了輕微的偽陽性結果增加。平衡技術（如 SMOTE）緩解了這種交易，特別是在高維度資料集（如 UNSW-NB15）中，若沒有平衡，偽陽性率會更高。

對抗訓練在對抗數據集上效果最為顯著，但在未修改數據集上影響有限。在這些情況下，SMOTE直接提升了召回率並減少了遺漏檢測。預處理步驟如規模化與特徵交互進一步提升了模型性能，而先進的特徵工程則提升了泛化能力，特別是對於複雜分類器。

微調提供了逐漸的性能提升，特別是在樹模型和深度學習模型中，通過優化對細微場景的性能。這些策略共同顯著提升了所有評估配置的準確率和召回率。

VII. 結論

在本文中，我們提出了一個綜合性的防禦框架，透過整合五個關鍵防禦策略——對抗訓練、平衡、特徵預處理、特徵聚合/工程以及模型微調——來增強基於機器學習的NIDS對抗攻擊的韌性。我們方法的一個關鍵方面是生成符合網絡協議約束和特徵相互依存性的真實對抗樣本。這些樣本不僅用於評估框架的穩健性，還通過對抗訓練來完善它，使模型能夠適應挑戰性的攻擊場景。對抗樣本生成與防禦策略之間的互動確保了框架能有效緩解演變中的威脅，使其實際適用於實際部署。

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TABLE II
PERFORMANCE ON NSL-KDD DATASET

Classifier Ensemble	Improvement Stage	Normal Accuracy	Normal Precision	Normal Recall	Adversarial Accuracy	Adversarial Precision	Adversarial Recall
TC Ensemble	Baseline	0.713	0.887	0.588	0.396	0.399	0.050
TC Ensemble	+ Adversarial Training & Balancing	0.865	0.915	0.832	0.742	0.792	0.644
TC Ensemble	+ Feature Engineering & Preprocessing	0.905	0.945	0.890	0.822	0.876	0.759
TC Ensemble	+ Model Fine-Tuning	0.931	0.967	0.915	0.920	0.952	0.910
DL Ensemble	Baseline	0.746	0.863	0.684	0.431	0.592	0.157
DL Ensemble	+ Adversarial Training & Balancing	0.894	0.920	0.853	0.815	0.859	0.722
DL Ensemble	+ Feature Engineering & Preprocessing	0.931	0.946	0.901	0.872	0.903	0.797
DL Ensemble	+ Model Fine-Tuning	0.960	0.953	0.982	0.940	0.933	0.969

TABLE III
PERFORMANCE ON UNSW-NB15 DATASET

Classifier Ensemble	Improvement Stage	Normal Accuracy	Normal Precision	Normal Recall	Adversarial Accuracy	Adversarial Precision	Adversarial Recall
TC Ensemble	Baseline	0.750	0.703	0.957	0.254	0.137	0.064
TC Ensemble	+ Adversarial Training & Balancing	0.860	0.813	0.972	0.684	0.579	0.512
TC Ensemble	+ Feature Engineering & Preprocessing	0.913	0.871	0.990	0.792	0.679	0.611
TC Ensemble	+ Model Fine-Tuning	0.944	0.912	0.996	0.942	0.916	0.987
DL Ensemble	Baseline	0.821	0.966	0.722	0.397	0.844	0.150
DL Ensemble	+ Adversarial Training & Balancing	0.885	0.939	0.805	0.743	0.877	0.689
DL Ensemble	+ Feature Engineering & Preprocessing	0.932	0.967	0.893	0.874	0.909	0.786
DL Ensemble	+ Model Fine-Tuning	0.977	0.993	0.969	0.963	0.981	0.956

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表 II 在 NSL-KDD 資料集上的性能

分類器集成	改善階段	正常準確率	正常精確率	正常召回率	對抗 對抗 准確率 精確度		對抗 召回率
TC 組合	基準	0.713	0.887	0.588	0.396	0.399	0.050
TC 組合	+ 對抗訓練與平衡	0.865	0.915	0.832	0.742	0.792	0.644
TC 組合	+ 特徵工程與預處理	0.905	0.945	0.890	0.822	0.876	0.759
TC 組合	+模型微調	0.931	0.967	0.915	0.920	0.952	0.910
DL 組合	基準	0.746	0.863	0.684	0.431	0.592	0.157
DL 組合	+ 對抗訓練與平衡	0.894	0.920	0.853	0.815	0.859	0.722
DL 組合	+ 特徵工程與預處理	0.931	0.946	0.901	0.872	0.903	0.797
DL 組合	+模型微調	0.960	0.953	0.982	0.940	0.933	0.969

表三 UNSW-NB15 數據集上的性能

分類器集成	改善階段	正常準確率	正常精確率	正常召回率	對抗準確率	對抗精確率	對抗召回率
TC 組合	基準	0.750	0.703	0.957	0.254	0.137	0.064
TC 組合	+ 對抗訓練與平衡	0.860	0.813	0.972	0.684	0.579	0.512
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DL 統合	+ 特徵工程與預處理	0.932	0.967	0.893	0.874	0.909	0.786
DL 統合	+模型微調	0.977	0.993	0.969	0.963	0.981	0.956

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